

Mastery Test 2 - Algorithms and Complexity

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1 Problem 1: Course Planning

Let X be the Course Planning problem. Let I_X be a YES Instance input to X . We'll define it as the tuple

$$I_X = (p, n, f, S, D, C, t)$$

where

- $p \in \mathbb{N}$ number of courses
- $n \in \mathbb{N}$ number of homeworks
- $f \in \mathbb{N}$ daily capacity
- $S = (s_1 \dots s_n) \in \mathbb{N}^n$ start dates
- $D = (d_1 \dots d_n) \in \mathbb{N}^n$ deadline dates
- $C = (c_1 \dots c_n) \in \{1 \dots p\}^n$ mapping of each homework to its course ID
- $t \in \mathbb{N}$ is the number of courses that need to be passed

1.1 Membership in NP

To verify a YES instance I_X we'll first define the certificate tuple

$$\varphi_x = (P, \sigma)$$

where

- $P \subseteq \{1 \dots p\}$ is the proposed set of courses that can be passed
- $\sigma = (\sigma_i)_{i \in H_P}$ is a schedule mapping $\sigma : H_P \rightarrow \mathbb{N}$
- σ_i is the day homework i is completed
- $H_P = \{i \in \{1 \dots n\} : c_i \in P\}$ is the set of homeworks required to pass the courses in p .

Then define the verification function which takes the instance and the certificate as inputs

$$V_X(I_X, \varphi_x)$$

where $V_X = 1$ (0 otherwise) iff the following conditions are met:

1. $t \leq |P|$ ensuring we passed t courses
2. $s_i \leq \sigma_i \leq d_i \forall i \in H_P$ ensuring all homework is completed in the correct time
3. $|\{x \in \sigma : x = k\}| \leq f \forall k \in \text{set}(\sigma)$ ensuring we do not go over capacity f on each day

The third check needs to search through σ for every element of σ so the # of checks is $H_P^2 \leq n^2$ i.e $V_X(\cdot)$ runs in $O(n^2)$. Verification runs in polynomial time $\Rightarrow X \in NP$

Algorithm 1 Verification Algorithm $V_X(I_X, \varphi_x)$

Require: Instance $I_X = (p, n, f, S, D, C, t)$ and Certificate $\varphi_x = (P, \sigma)$

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1:  $H_P \leftarrow \{i \in \{1 \dots n\} : c_i \in P\}$ 
2: if  $t > |P|$  then
3:   return 0
4: end if
5: for each  $i \in H_P$  do
6:   if  $\sigma_i < s_i$  or  $\sigma_i > d_i$  then
7:     return 0
8:   end if
9: end for
10: for each  $k \in \text{set}(\sigma)$  do
11:    $count \leftarrow |\{x \in \sigma : x = k\}|$ 
12:   if  $count > f$  then
13:     return 0
14:   end if
15: end for
16: return 1
```

1.2 NP-Hardness Reduction

We'll show that X (Course Planning) is NP-hard by reducing a different NP-Complete problem Y . We have to show that this reduction function $R(\cdot) : I_Y \rightarrow I_X$ runs in polynomial time.

We'll choose Independent Set as problem Y . The YES Instance I_Y will be the following tuple

$$I_Y = (G, k)$$

made of a graph $G = (V, E)$ and integer $k \in \mathbb{N}$. The reduction function's output will need to be in the form of the tuple

$$R(I_Y) = (p, n, f, S, D, C, t)$$

Here is my proposal for how to set the values for the output of this function:

1. For simplicity set the daily capacity to $f = 1$
2. Make $p = |V|$ courses, each corresponding to a vertex $v \in V$.
3. Set the target number of passed courses to $t = k$.
4. Begin with $i \leftarrow 1$
 $\forall e = \{v, u\} \in E$:
 - (a) create unique day $d_e \in \mathbb{N}$
 - (b) Homework i will have course index $c_i = v$, with start date and deadline as $s_i = d_i = d_e$
 - (c) Homework $i + 1$ will have course index $c_{i+1} = u$, with start date and deadline as $s_{i+1} = d_{i+1} = d_e$
 - (d) Insert these values of c_i, s_i, d_i in to S, D, C
 - (e) increment the homework index ($i \leftarrow i + 2$)
5. The total number of homeworks will be $n = 2|E|$.

Algorithm 2 Reduction Algorithm $R(I_Y)$

Require: Independent Set Instance $I_Y = (G, k)$ where $G = (V, E)$

```
1:  $f \leftarrow 1$ 
2:  $p \leftarrow |V|$ 
3:  $t \leftarrow k$ 
4:  $n \leftarrow 2|E|$ 
5: Initialize  $S, D, C$  as empty arrays
6:  $i \leftarrow 1$ 
7: for each edge  $e = \{u, v\} \in E$  do
8:   Create unique day  $d_e \in \mathbb{N}$ 
9:    $c_i \leftarrow v$ 
10:   $s_i \leftarrow d_e$ 
11:   $d_i \leftarrow d_e$ 
12:   $c_{i+1} \leftarrow u$ 
13:   $s_{i+1} \leftarrow d_e$ 
14:   $d_{i+1} \leftarrow d_e$ 
15:  Append  $(s_i, s_{i+1})$  to  $S$ 
16:  Append  $(d_i, d_{i+1})$  to  $D$ 
17:  Append  $(c_i, c_{i+1})$  to  $C$ 
18:   $i \leftarrow i + 2$ 
19: end for
20: return  $I_X = (p, n, f, S, D, C, t)$ 
```

Time Complexity of Reduction: In step 1 we needed to iterate through all the vertices to assign course indices ($|V|$ operations required), to then assign the homeworks to each course takes $2|E|$ operations in step 4. So in total the reduction $R(\cdot)$ runs in $O(|V| + |E|)$ time which is polynomial.

1.3 Correctness Proof

($\Rightarrow$) YES instances map to YES instances: Suppose $I_Y = (G, k)$ is a YES instance. This implies $\exists V' \subseteq V$ s.t that $k \leq |V'|$. Which constructs the certificate for Y as $\varphi_y = G' = (V', E_{G'})$. We can use this to construct a valid certificate

$$\varphi_x = (P, \sigma)$$

corresponding to

$$I_X = R(I_Y) = (|V|, 2|E|, 1, S, D, C, k)$$

the following way:

1. Let $P = V'$ which satisfies the first condition $t \leq |V'|$ since $t = k$
2. The set of required homework $H_P = \{i \in \{1 \dots n\} : c_i \in P\}$. For all $i \in H_P$, the schedule must be set to $\sigma_i = d_i$ because $s_i = d_i = d_e$ which satisfies the second condition $s_i \leq \sigma_i \leq d_i$
3. Now that we constructed the cert., we have to verify the third condition that $f = 1$ is not exceeded. By the reduction, each unique day d_e corresponds to one edge $e = (v, u) \in E$. There's only 2 homeworks that can be done on this day and if $v \in V'$ then $u \notin V'$ isn't in the Independent Set. Which guarantees only at most one homework will be completed on any day d_e

Because all verification conditions are met I_Y is YES instance $\Rightarrow R(I_Y)$ is YES instance. $\square$

($\Leftarrow$) **YES instances map to YES instances:** Now we have to prove the other direction that if a valid cert. for $R(I_Y)$ exists that a valid cert. for I_Y exists.

1. Assume $I_X = R(I_Y)$ is a YES instance which implies $\exists \varphi_x = (P, \sigma)$ s.t $V_X(I_X, \varphi_x) = 1$
2. Construct $V' = \{v \in V : v \in P\}$ since each course index is unique to each vertex $\Rightarrow k \leq |V'| = |P|$ (from condition #1 in the verifier V_X)
3. Take any 2 distinct vertices $v, u \in V' \Leftrightarrow v, u \in P$
4. Note that the homeworks in v, u are present in H_P . Lets make an assumption that there exists an edge $\exists e = (v, u) \in E$.
5. This implies that v, u have each have a homework with the same deadline $s_i = d_i = d_e$
6. Due to condition # 3 and because φ_x is valid, we know that the # of completed homeworks don't exceed $f = 1$ on any day σ_i . Statement 4 contradicts this so our assumption must be false.
7. This implies that if you select any 2 distinct vertices from V' the edge $e = (v, u)$ does not exist. Thereby V' is by an independent set. $\square$